

can press Ctrl+F4, click on Exit in the File menu, or click the Exit button provided at the bottom right of the software. In the Tools menu, you can access the system tools like calculator, notepad, clipboard and volume control. Details about the pin-out diagram of the serial port can be found in the Help menu.

When you start the application, the software automatically checks the availability of serial ports on the computer and displays a list so that it becomes easy to get started with the known available serial port. The user can select a particular serial port from the drop-down menu list. The button to the left of this drop-down menu is for opening or closing the serial port communication. On the right side of the screen, you can also see a box where four tabs are given to set the communication configurations, viz, General, Buffer, Hardware and Other.

As shown in Fig. 4, the first settings tab is the General tab. Here you can select the baud rate from 110 to 25,600. Parity settings include even/odd/none/mark/spaces. With the number of bits list, you can select the bits amongst 4/5/6/7/8. With the number of stop bits list, you can select amongst 1/1.5/2. The handshaking refers to the internal communication protocol by which the data is transferred from the hardware port to receive buffer. It ensures that the data is not lost due to a buffer overrun, where data arrives at the port too quickly for the communication device to move the data in the receive buffer. The handshaking menu allows you to choose from none/XonXoff/RTS-CTS/BOTH. The default setting is shown in Fig. 4.

As shown in Fig. 5, the second settings tab is the Buffer tab. Here you will find some advanced settings for the serial reception. It is advised not to configure these settings until you really know what you are doing. As the software uses 'MSCOMM32.ocx' ActiveX object to establish communication with the computer's serial port, this tab basically defines the advanced setting for the object to communicate.

Fig. 4: Default values under general tab

Fig. 5: Buffer settings tab

Fig. 6: Hardware settings tab

Fig. 7: Other settings tab

With in/out buffer size (in number of bytes), you can specify the computer's internal buffers size for input/output of the mscomm32.ocx object. RThreshold sets the number of characters to receive before the MSCOMM control sets the CommEvent property to comEvReceive and generates the OnComm event. SThreshold sets the minimum number of allowable characters in the transmit buffer before the MSCOMM control sets the CommEvent property to comEvSend and generates the OnComm event. Inputlen sets the number of characters the Input property reads from the receive buffer.

As shown in Fig. 6, the third settings tab is the Hardware tab. It allows you to control the serial hardware pins of the port such as DTR and RTS. The Parity Replace option sets the character that replaces an invalid character in the data stream when a parity error occurs.

The Null Discard determines whether null characters are transferred from the port to the receive buffer. The DTR Enable determines whether to enable the data terminal ready (DTR) line during communication. Typically, the DTR signal is sent by a computer to its modem to indicate that the computer is ready to accept incoming transmission. With the default button, you can reset the software settings to default. The RTS Enable determines whether to enable the RTS line. Typically, the RTS signal that requests permission to transmit data is sent from a computer to its attached modem.

As shown in Fig. 7, the fourth settings tab is the Other tab. Here you can alter different feature settings of the software. If the 'Time logging on reception' is checked, the software will also log the date and time on each received serial data so that the user can have an exact idea that which data came at what time. If the 'Clean receive every data' is checked, the software will wash the previously received data from the Receive box to update the latest one. If the 'Speak everytime on reception' is checked, the software will speak the message each time it receives the data. The normal GUI interaction sounds can be enabled by checking the 'Enable GUI Sounds' check box. If 'Flash receivebox on reception' is checked, the software will flash the Receive box each time it receives new data.

Functions and routines. The software uses a couple of functions and routines to break a bigger problem into smaller modules and build a crash-proof, robust and easy-to-use application. The software contains the following main event handlers and routines:

Form_Load(). This is the code